



EDUCATION

Year	Degree/Exam	Institute	CGPA/Marks
2027	4YRS B.S	IIT Kharagpur	8.43 / 10
2023	Central Board of Secondary Education (CBSE)	Kendriya Vidyalaya IIT Kharagpur	91.2%
2021	Central Board of Secondary Education (CBSE)	Kendriya Vidyalaya IIT Kharagpur	97.6%

INTERNSHIPS

Data Science Intern | Polestar Analytics

May 2026 - Jul 2026

Project: Agentic Stock Manager | Human-in-the-loop system for automated supply-chain replenishment

- Built a **multi-agent system** to detect stock threshold breaches and generate **purchase orders** from live inventory, warehouse, and product data.
- Integrated real-time APIs, multi-LLM routing, purchase order approvals, guardrails, and audit logs into a **6-agent LangGraph workflow**.
- Consolidated per-SKU orders into **one purchase order per vendor** using vendor based batching, supporting auto issue, draft, batch, and suppress

Project: Promotional Uplift Modeling | Time-aware forecasting and causal promotion targeting

- Built a time aware demand forecasting pipeline integrating multi-source retail data with lag, rolling, trend, and **Fourier features**.
- Developed and benchmarked **CatBoost T-, S-, and X-learners** alongside hierarchical Bayesian uplift models using **Qini, AUUC, uplift@K**
- Converted uplift estimates into a capacity constrained **promotion policy**, prioritizing product-weeks by predicted incremental demand

Analyst Programmer Intern | Xceedance Consulting Pvt. Ltd.

May 2025 - Sept 2025

- Architected an end to end insurance document pipeline using **Tesseract OCR** for text extraction and **BERT** based NER for entity extraction
- Fine tuned **BERT** to extract insurance entities from OCR text, achieving **94.2%** accuracy and **0.91** weighted **F1**, reducing costly LLM API calls.
- Implemented **RapidFuzz** based entity **normalization** to map noisy NER outputs to standardized reference records, reducing manual review
- Built a **Streamlit** interface for NER and normalization, integrating **MLflow** tracking and automated **Databricks** ML pipeline execution.

Undergraduate Student Researcher | Prof. Pawan Goyal | CNeRG Lab, Department of Computer Science and Engineering, IIT Kharagpur

- Scraped and curated Sanskrit text from Rigveda and Atharvaveda using **BeautifulSoup**, aligning svara verses with audio for **ASR** dataset
- Fine-tuned **Wav2Vec2** and **IndicWhisper** by expanding **vocabularies** to accommodate **Vedic svaras** for Sanskrit speech recognition.
- Evaluated speech-to-text transcription using **Word Error Rate (WER)** and **Character Error Rate (CER)**, achieving **17.5%** WER and **6.2%** CER.

PROJECTS

Academic Outreach AI | Agentic LLM &Automation Pipeline | Self Project

- Built an evidence-first **LangGraph** pipeline for personalized faculty outreach, reducing generation from 48 LLM calls to 2 calls/email.
- Engineered **parallel RAG** retrieval using **Semantic Scholar**, faculty websites, Tavily, and async crawlers with **SQLite TTL** caching.
- Implemented hallucination resistant generation with **Pydantic** schemas, evidence constraints, retries, fallbacks, and human approval.
- Developed a **Streamlit** platform with async concurrency, approval workflows, Gmail API, OAuth refresh, and secure **MIME** encoding.

Market Regime Modeling and Adaptive Strategy Selection | Self Project

- Built a Market regime platform comparing **HSMM, BOCPD, GMM, and XGBoost** using point-in-time price and macroeconomic features.
- Implemented causal **HSMM** filtering with online posterior estimation and mapped latent states to 5 interpretable market regimes.
- Developed a probabilistic strategy allocator using regime probabilities, confidence-based selection, and risk-aware portfolio scaling.
- Evaluated models with purged **walk-forward validation** and transaction costs; achieved **1.61 Sharpe** vs. 0.84 benchmark with 9.0% Max DD.

Fan-Out MCP: Multi-Agent Delegation Plugin | Self Project

- Architected a cross-harness **MCP** runtime for **Claude Code**, Codex and Copilot CLI, dispatching 5-30 subtasks to local and hosted SLMs.
- Implemented an async execution engine with bounded **concurrency**, timeouts, exponential backoff, and cost ceilings for reliable execution.
- Engineered **multi-strategy** aggregation using majority voting, LLM-as-Judge scoring, and Map-Reduce with prompt injection isolation.
- Designed a local-first, default-deny provider framework with pluggable routing, **OpenTelemetry** tracing, and **JSONL** token accounting.

COMPETITION/CONFERENCE

- Achieved Codeforces **Expert** (peak 1606; best rank 141 in Div. 2) and solved **1000+ DSA problems** across LeetCode, CSES and other platforms.
- Ranked top **50** on the final leaderboard and top 10 on the rolling leaderboard in IIT Pokerbots. Title Sponsor: Jane Street, Organizer: IITM
- Secured **4th** place in Open IIT Data Analytics 2024 by building a supervised learning model with feature engineering to predict flight delays.
- Participated in multiple **Kaggle** ML competitions, applying preprocessing, feature engineering, modeling, and evaluation.

SKILLS AND EXPERTISE

Programming Languages and Tools: *Proficient:* C/C++, Python, SQL, Git, Bash

Familiar: HTML, CSS, JS, R

Frameworks and Libraries: NumPy, Pandas, Matplotlib, Seaborn, Scikit-Learn, CatBoost, XGBoost, LightGBM, Random Forest, Statsmodels, PyTorch, Optuna, Transformers, MLFlow, BeautifulSoup, LangGraph, FastAPI, Streamlit, RapidFuzz, Pydantic, AsyncIO, SQLAlchemy, YAML

Tools: VS Code, Databricks, Jupyter Notebook, n8n, MS Office, Linux (Ubuntu), Github, Anaconda, Kaggle, Matlab (Familiar), Vercel

COURSEWORK INFORMATION

Computer Science: Information Retrieval, Machine Learning, Algorithms (Theory & Laboratory) , Programming & Data Structures (Theory & Laboratory) , Operating Systems, Database Management Systems, Deep Learning, Computer Networks, Natural Language Processing

Maths: Linear Algebra & Numerical Analysis, Advanced Calculus, Probability & Statistics, Partial Differential Equations, Transform Calculus

Core: Methods of Prospecting, Pattern Recognition, Borehole Geophysics, Geophysical Signal Processing, Geophysical Inverse & Field Theory